Building Design System for Suarise Differently-Abled Users, Providing Accessible Experience for Everyone

Design System Project

Project overview

The problem

Suarise is an Indonesian company focused on digital accessibility. They wanted to build a platform where differently-abled users could report accessibility issues on websites and apps, and developers could learn how to fix them.

During early reviews, I found that many users struggled with basic usability, such as unclear buttons, low-contrast text, and layouts that screen readers couldn't interpret. It became clear that we couldn't move forward without first creating a design system grounded in accessibility standards.

The goal

Create a scalable, WCAG-compliant design system to make the platform inclusive and consistent for everyone.

My role

I was the sole UX designer, working with the director, UX researcher, UX writer, and frontend engineers. I handled:

- User and competitor research
- Accessibility standards and design principles
- Component design and documentation
- Prototyping and testing with differently-abled users

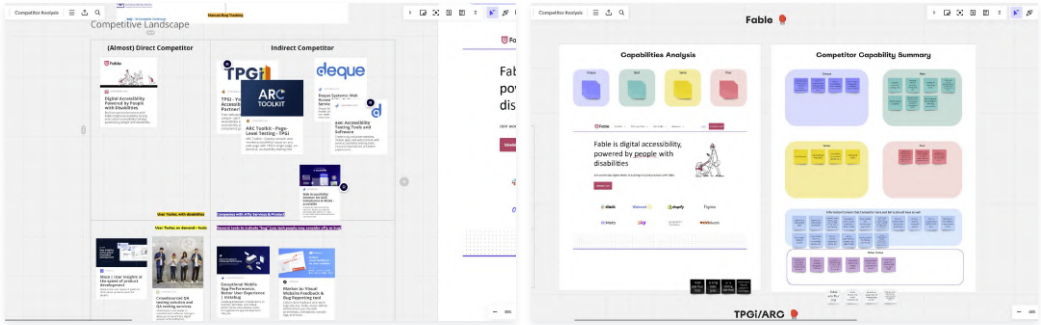
Duration

1 month

Design study

Competitor Analysis

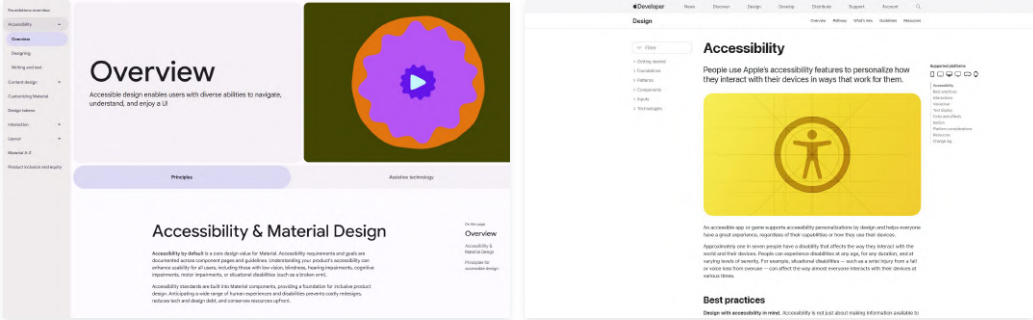
I studied accessibility-focused tools such as ARC Toolkit and axe DevTools to understand how they designed for feedback and reporting. Most were very technical and not friendly for non-developers, so I aimed to create something approachable yet fully compliant.



Studying competitors to learn how accessibility tools support differently-abled users.

Design Systems Research

I explored Google's Material Design and Apple's Human Interface Guidelines to balance accessibility and aesthetics. Material excelled in scalability and contrast, while Apple focused on clarity and hierarchy. I combined both philosophies to create a design system that felt modern, structured, and easy to use.



Reviewing Google and Apple design systems to identify accessible patterns and reusable components.

Build the design system

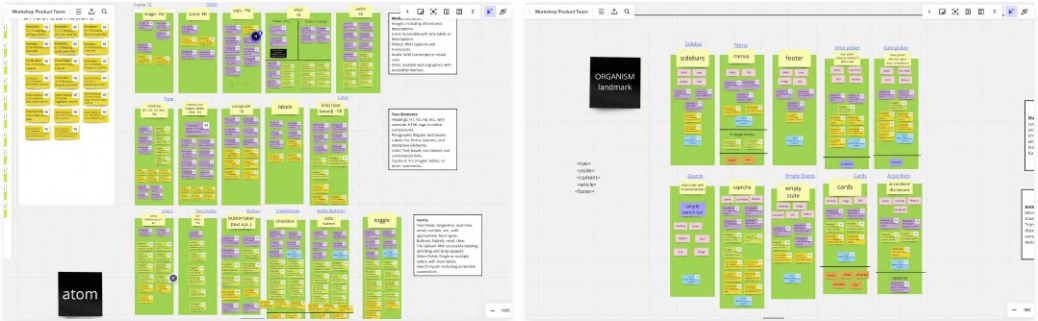
Accessibility Standards Check

I audited buttons, forms, text, and navigation, finding many that failed WCAG standards for contrast and hierarchy. I created a checklist to evaluate each element, helping the team identify what needed improvement.

Built the Design System

I designed a complete system that included:

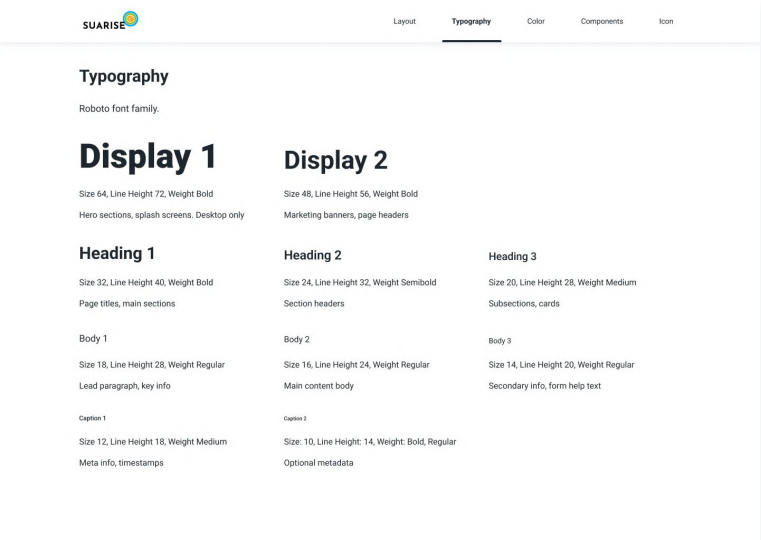
- **Typography**: Clear hierarchy with readable type scales
- **Color palette**: High-contrast colors tested with Stark
- **Icons**: Simple shapes readable at any size
- **Components**: Reusable elements with focus and hover states, plus ARIA labels



Auditing all UI elements against WCAG criteria to ensure compliance.

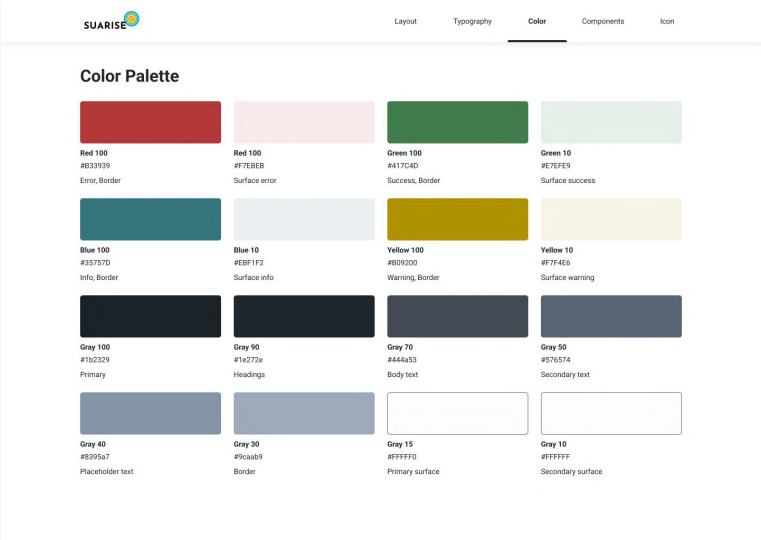
Build the design system

Defining clear text styles and hierarchy for easy reading.



Typography

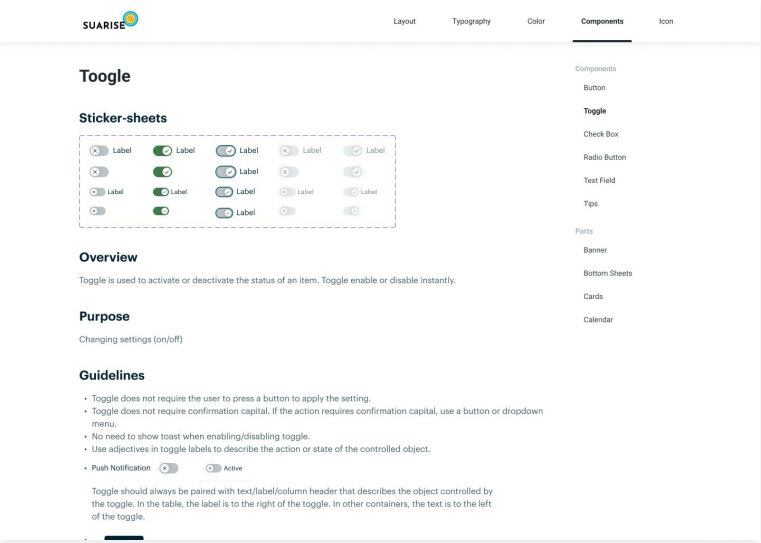
Setting contrast-safe color rules based on WCAG standards.



Color

Build the design system

Documenting reusable elements with accessibility notes and usage guides.



Component

Build the design system

Collaboration and Testing

When I presented the first version, engineers requested real-world validation beyond compliance checklists. We built a sign-up flow prototype and tested it with differently-abled users. Early feedback showed three issues:

- Too many form fields
- Screen readers couldn’t read nested labels
- Too much content on one page

I worked with engineers to simplify the layout, improve navigation, and fix ARIA labeling. After two testing rounds, users completed the sign-up faster and made fewer mistakes.

Testing user onboarding with proper label and navigation support for screen readers.

Takeaways

The result

The new design system made accessibility a foundation of Suarise's product. It standardized fonts, colors, icons, and 24 components, helping the team design and build faster while ensuring inclusivity.

What I learned

Working with differently-abled users taught me that accessibility isn't just about rules; it's about real people. I learned to test early, listen carefully, and design with empathy as much as precision. If I were to do it again, I would introduce accessibility testing earlier and strengthen the feedback loop between design and engineering.